Alternative Rotation Representations

ECE/MAE 5930 Mobile Robotics

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2D Rotations: Unit Complex Numbers

- Angles are great, but not unique (Restric ourselves to $[0, 2\pi)$ or $[-\pi, \pi)$)

  $\theta = \theta + 2\pi k$ where $k \in \mathbb{R}$

  Unique

- Matrices are easy and unique
  - Nice: part of SO(2)
  - Not nice: 4 parameters

$$R(\theta) = \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix}$$

- Unit complex numbers
  - Imaginary numbers represent angles in polar coordinates

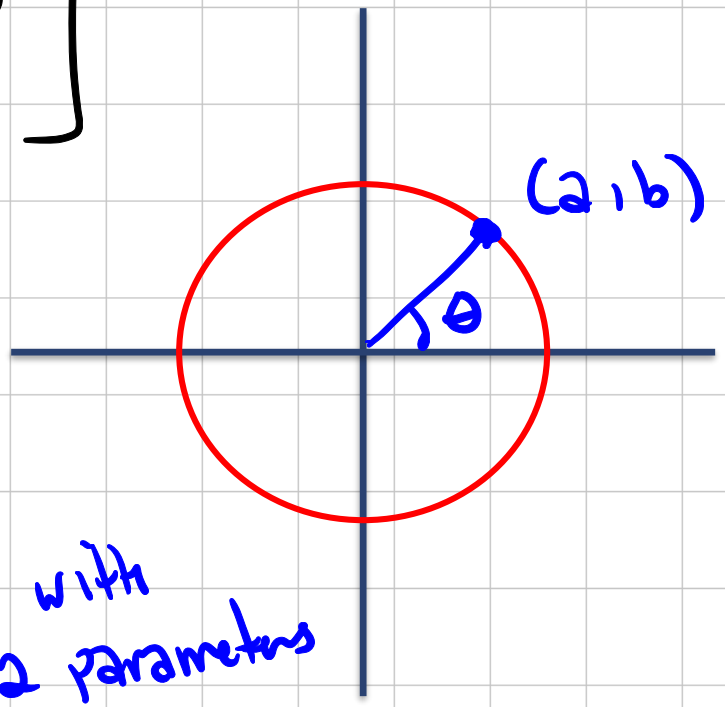
    $re^{i\theta}$ where $r=1$ (Unit circle)

  - Conversion to cartesian coordinates

    $a+bi$ $\quad a = r\cos\theta,\ b = r\sin\theta$

  - Unit

    $a = \cos\theta \quad r = \sin\theta$



Unique with only 2 parameters

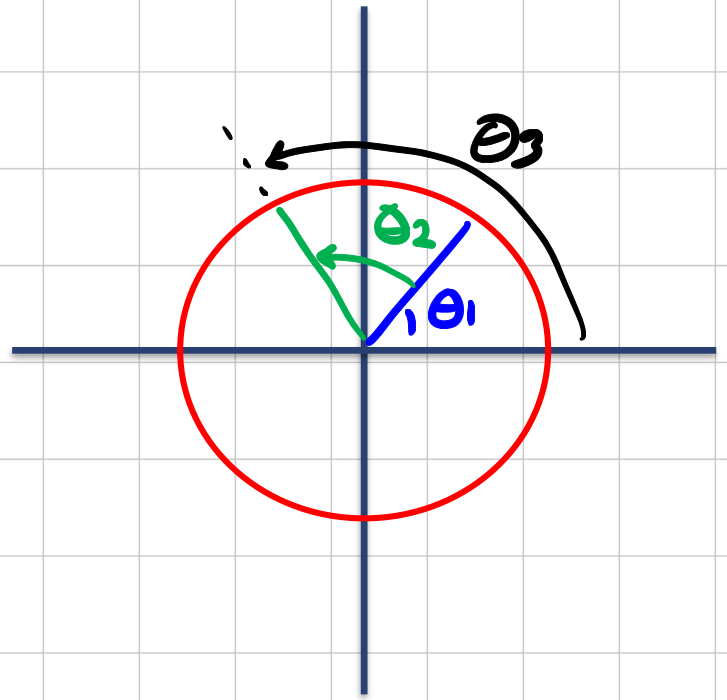
Unit Complex Numbers: Combining Rotations

- Polar form

$$e^{i\theta_1} \cdot e^{i\theta_2} = e^{i\theta_3}$$

- Cartesian form

$$a_3 = a_1 a_2 - b_1 b_2$$
$$b_3 = a_1 b_2 + a_2 b_1$$



- Interesting note:
  - All rotations in 2D or 3D are about a single axis!
  - What axis are 2D rotation rotating about? z

3D Rotations – Euler Coordinates

- Fact: Any 3D rotation can be expressed using the Elementary Direction Cosine matrices (i.e. they form a basis)

- Example: Euler X-Y-Z (roll, pitch, yaw)

$$R(\gamma,\beta,\alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

  - Which rotation is performed first?
  - Extracting roll-pitch-yaw

Note error in book

$$\begin{cases} \alpha = atan2(r_{21}, r_{11}) \\ \beta = atan2(-r_{31}, \sqrt{r_{32}^2 + r_{33}^2}) \\ \gamma = atan2(r_{32}, r_{33}) \end{cases}$$

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

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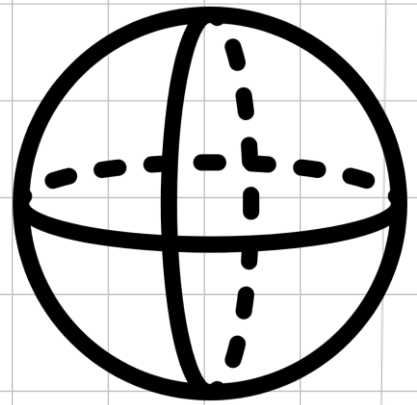
3D Rotations - Quaterions

- Angles are not a unique representation (Singularities exist in Euler coordinates)
- Rotation matrix: 9 parameters to represent 3 DOF
- Quaterions – Generalization of polar coordinates (almost)
  - Idea: any rotation described by axis and rotation angle
  - 3 variables: axis
  - 1 variable: rotation
- Properties



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Combining Quaterions

- Multiplying two angles

$$
\begin{aligned}
a_3 &= a_1 a_2 - b_1 b_2 - c_1 c_2 - d_1 d_2 \\
b_3 &= a_1 b_2 + a_2 b_1 + c_1 d_2 - c_2 d_1 \\
c_3 &= a_1 c_2 + a_2 c_1 + b_2 d_1 - b_1 d_2 \\
d_3 &= a_1 d_2 + a_2 d_1 + b_1 c_2 - b_2 c_1 .
\end{aligned}
\tag{4.19}
$$

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3D Rotations – Matrices and Quaternions

- Calculating matrix from quaternion

$$R(h) = \begin{pmatrix} 2(a^2+b^2)-1 & 2(bc-ad) & 2(bd+ac) \\ 2(bc+ad) & 2(a^2+c^2)-1 & 2(cd-ab) \\ 2(bd-ac) & 2(cd+ab) & 2(a^2+d^2)-1 \end{pmatrix}, \tag{4.20}$$

$$a = \tfrac{1}{2}\sqrt{r_{11}+r_{22}+r_{33}+1}, \tag{4.24}$$

$a \neq 0$

$$b = \frac{r_{32}-r_{23}}{4a}, \tag{4.25}$$

$$c = \frac{r_{13}-r_{31}}{4a}, \tag{4.26}$$

$$d = \frac{r_{21}-r_{12}}{4a}. \tag{4.27}$$

$a = 0$

$$b = \frac{r_{13}r_{12}}{\sqrt{r_{12}^2 r_{13}^2 + r_{12}^2 r_{23}^2 + r_{13}^2 r_{23}^2}},$$

$$c = \frac{r_{12}r_{23}}{\sqrt{r_{12}^2 r_{13}^2 + r_{12}^2 r_{23}^2 + r_{13}^2 r_{23}^2}},$$

$$d = \frac{r_{13}r_{23}}{\sqrt{r_{12}^2 r_{13}^2 + r_{12}^2 r_{23}^2 + r_{13}^2 r_{23}^2}}.$$

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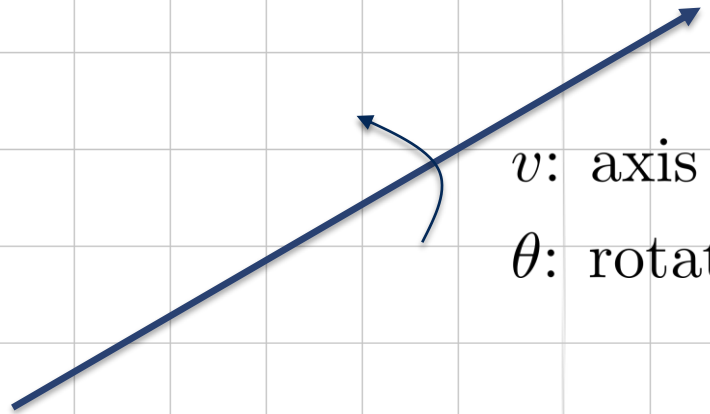
Summary

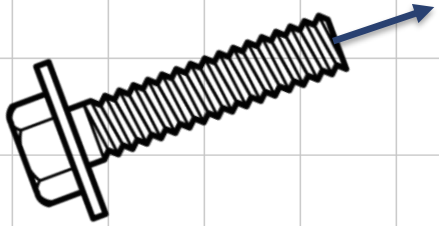
- 2D Rotations
  - Angles: not unique, 1 parameter, 1 dof
  - Unit complex number (spherical set of 1 dimension): unique, 2 parameters, 1 dof
  - Rotation matrix: unique, 4 parameters, 1 dof
- 3D Rotations
  - Euler angles: not unique, 3 parameters, 3 dof
  - Quaternions: two possibilities (not unique), 4 parameters, 3 dof
  - Rotation matrix: unique, 9 parameters, 3 dof
- Each rotation can be represented about a single axis



v: axis of rotation

θ: rotation amount

Fact (Screw theory): A single axis can be used to describe any transform (i.e. a rotation and translation about the axis)



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